

cell to be equal to the minimum or maximum possible value. However, the associated variability of the MLE varies across models. For example, in some cases broadly similar MLEs are obtained, as for model $\{12, 13, 14, 34, 134\}$; whereas for other models, such as model $\{12, 13, 23, 14, 34\}$ the MLEs are very different. Further, we note that the model deemed optimal differs for the case where all cell values are set to their maximum value, compared to the other two approaches.

Table 1 Contingency table for the number of injecting drug users for the North East region for young females. An * indicates that the cell entry is interval censored and lies within the range 1-4.

S_1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
S_2	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
S_3	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
S_4	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	74	17	*	584	63	12	9	35	*	*	*	35	17	*	*

Table 2 MLEs and 95% CIs of the total population size for the models identified; and associated ΔAIC ($= AIC - \min(AIC)$) and ΔBIC ($= BIC - \min(BIC)$) statistics for each of the different methods applied to the censored contingency table data. The model is specified via the maximum interaction terms present; and the model deemed optimal is highlighted in bold for each method.

Method	Model	MLE	95% CI	ΔAIC	ΔBIC
Cells = minimum value	$\{12, 13, 14, 34\}$	2107	(1496, 3567)	6.86	6.14
	$\{12, 13, 23, 14, 34\}$	4541	(1998, 7126)	0	0
	$\{12, 13, 14, 34, 134\}$	1648	(1238, 2319)	2.85	2.84
	$\{12, 13, 23, 14, 34, 134\}$	3179	(1232, 4981)	1.39	2.09
	$\{12, 13, 23, 14, 34, 123\}$	4206	(1662, 6592)	2.00	2.69
	$\{12, 13, 23, 14, 34, 123, 134\}$	1390	(1071, 1692)	1.81	3.22
Cells = maximum value	$\{12, 13, 14, 34\}$	1642	(1236, 1801)	3.25	1.12
	$\{12, 13, 23, 14, 34\}$	1794	(1180, 1814)	4.67	3.25
	$\{12, 13, 14, 34, 134\}$	1417	(1071, 1560)	1.87	0.45
	$\{12, 13, 23, 14, 34, 134\}$	1331	(973, 1376)	3.79	3.07
	$\{12, 13, 23, 14, 34, 123\}$	1646	(1066, 1679)	6.30	5.59
	$\{12, 13, 23, 14, 34, 123, 134\}$	1010	(908, 1084)	0	0
Cells = interval	$\{12, 13, 14, 34\}$	1933	(1379, 2636)	2.46	1.75
	$\{12, 13, 23, 14, 34\}$	3060	(1583, 3858)	0	0
	$\{12, 13, 14, 34, 134\}$	1597	(1178, 2273)	0.82	0.82
	$\{12, 13, 23, 14, 34, 134\}$	2293	(1104, 3456)	1.40	2.11
	$\{12, 13, 23, 14, 34, 123\}$	2970	(1417, 4073)	1.99	2.70
	$\{12, 13, 23, 14, 34, 123, 134\}$	1113	(924, 1282)	1.32	2.73

5 Discussion

We have described an approach for dealing with general interval censored data within a multiple systems estimation framework, taking into account the uncertainty of the censored contingency table cells. Such data may arise, for example, due to privacy issues, where small cell counts are not published exactly, but instead an interval is given for the cell count. Further we have explored the sensitivity of the population size estimates in relation to how these censored cells are dealt